Model Research Document

AI-Powered Workforce Analytics & Talent Intelligence Dashboard

1. Project Overview

The dashboard applies AI/ML techniques to workforce data to generate actionable insights on employee attrition risk, performance trends, skill gaps, and engagement — enabling data-driven HR and management decisions. Since a single model cannot address every requirement, the system is decomposed into distinct ML sub-problems, each paired with the most suitable modeling approach based on data availability, accuracy needs, and explainability requirements.

2. Key Objectives

- Predict employee attrition risk and forecast workforce/hiring demand.
- Analyze performance trends and segment talent into actionable groups.
- Extract skills from resumes/profiles and identify skill gaps against roles.
- Gauge employee sentiment/engagement from feedback text.
- Recommend upskilling, training, or internal mobility actions.

3. Sub-Problems & Recommended Models

Sub-Problem	Task Type	Recommended Model	Justification
Attrition Prediction	Classification	Random Forest / XGBoost	High accuracy on tabular HR data; explainable via SHAP
Performance Prediction	Regression	XGBoost / CatBoost	Handles mixed categorical + numerical KPI data well
Demand Forecasting	Time-Series	Facebook Prophet	Interpretable, handles seasonality, works with limited history
Skill Extraction	NLP - NER	spaCy (custom NER)	Good accuracy-to-effort ratio for resume parsing
Skill Gap Analysis	Similarity Matching	Sentence-BERT + Cosine Similarity	Captures semantic closeness (e.g., ML ≈ Machine Learning)
Sentiment/ Engagement	NLP - Text Classification	Pre-trained DistilBERT	High accuracy, fast integration, minimal labeled data needed
Talent Segmentation	Clustering	K-Means + DBSCAN	Interpretable clusters; DBSCAN flags outlier/at-risk profiles
Recommendations	Recommender System	Content-Based Filtering	Effective without large interaction/history datasets

4. Proposed Technology Stack

Layer	Tools / Technologies
Data Processing	Python, Pandas, NumPy
ML Modeling	Scikit-learn, XGBoost, CatBoost, Prophet
NLP	spaCy, Hugging Face Transformers, Sentence-BERT
Visualization / Dashboard	Power BI / Tableau, or Plotly Dash / Streamlit
Backend	Flask / FastAPI
Storage	PostgreSQL / CSV / Cloud Storage

5. Evaluation Metrics

Task Type	Metrics
Classification (Attrition, Sentiment)	Accuracy, Precision, Recall, F1-Score, ROC-AUC
Regression (Performance)	RMSE, MAE, R^2
Forecasting (Demand)	MAPE, RMSE

Task Type	Metrics
Clustering (Segmentation)	Silhouette Score, Davies-Bouldin Index
NLP Extraction (Skills/NER)	Precision, Recall, F1 (entity-level)
Recommendations	Precision@K, Recall@K

6. Risks & Considerations

- Data Privacy: Employee data is sensitive — anonymization and access control are essential.
- Bias: Attrition/performance models must be audited for bias across gender, age, or department.
- Explainability: HR stakeholders need interpretable outputs — prioritize SHAP/LIME over black-box scores.
- Data Volume: Limited historical data favors classical ML over deep learning in early phases.

7. Conclusion

This research establishes a modular modeling strategy for the AI-Powered Workforce Analytics & Talent Intelligence Dashboard, pairing each sub-problem with a feasible, well-justified model. The recommended stack balances predictive accuracy, explainability, and practical implementation feasibility within an internship project timeline.